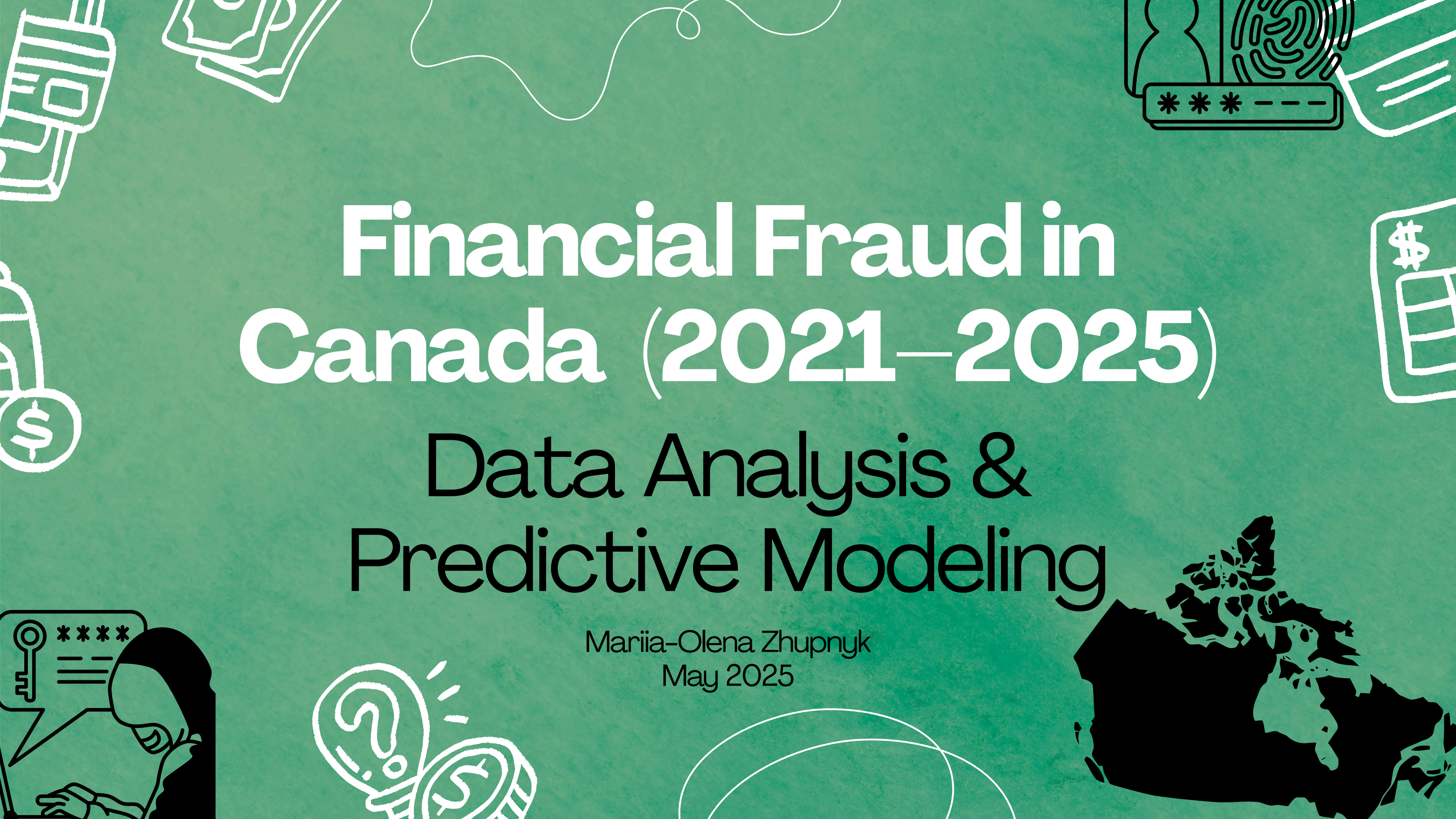


Financial Fraud in Canada (2021–2025)

Data Analysis & Predictive Modeling

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Project Objective

- Understand patterns in fraud complaints
- Identify high-risk groups and fraud methods
- Analyze financial loss trends
- Predict likelihood of financial loss from complaints

Why It Matters:

Fraud affects thousands of Canadians annually—this analysis helps reveal where and when it's happening most.



Data Source & Tools

- Source: Government of Canada - Open Data
- Period Covered: Jan 2021 – Mar 2025
- Scope: Fraud type, reported losses, dates, location, demographics
- Tools: Python (Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, Plotly)
- Jupyter Notebook, GoogleColab

```
1 # Preview data
2 print("Initial shape:", df.shape)
3 df.head()
```

Initial shape: (328649, 21)

	Número d'identification / Number ID	Date Received / Date reçue	Complaint Received Type	Type de plainte reçue	Country	Pays	Province/State
0	1	2021-01-02	CAFC Website	CAFC site web	Canada	Canada	Saskatchewan
1	2	2021-01-02	CAFC Website	CAFC site web	Not Specified	Non spécifié	Not Specified
2	3	2021-01-02	CAFC Website	CAFC site web	Canada	Canada	Quebec
3	4	2021-01-02	CAFC Website	CAFC site web	Canada	Canada	Saskatchewan
4	5	2021-01-02	CAFC Website	CAFC site web	Canada	Canada	Saskatchewan

5 rows × 21 columns

Data Cleaning & Preparation

- Remove Duplicate Columns (Keep English only)
- Rename columns for clarity,
- Removing extra characters and whitespace,
- Converting Dollar_Loss to numeric,
- Converting Date_Received to the correct format
- Standardizing age ranges and other categorical values,
- Handling missing or unspecified entries properly.

```
2 df = df.replace(['Not Available', 'non disponible', 'Not Specified', 'Autre/inconn
```

```
1 # Check missing values
2 missing_summary = df.isna().sum()
3 missing_summary
```

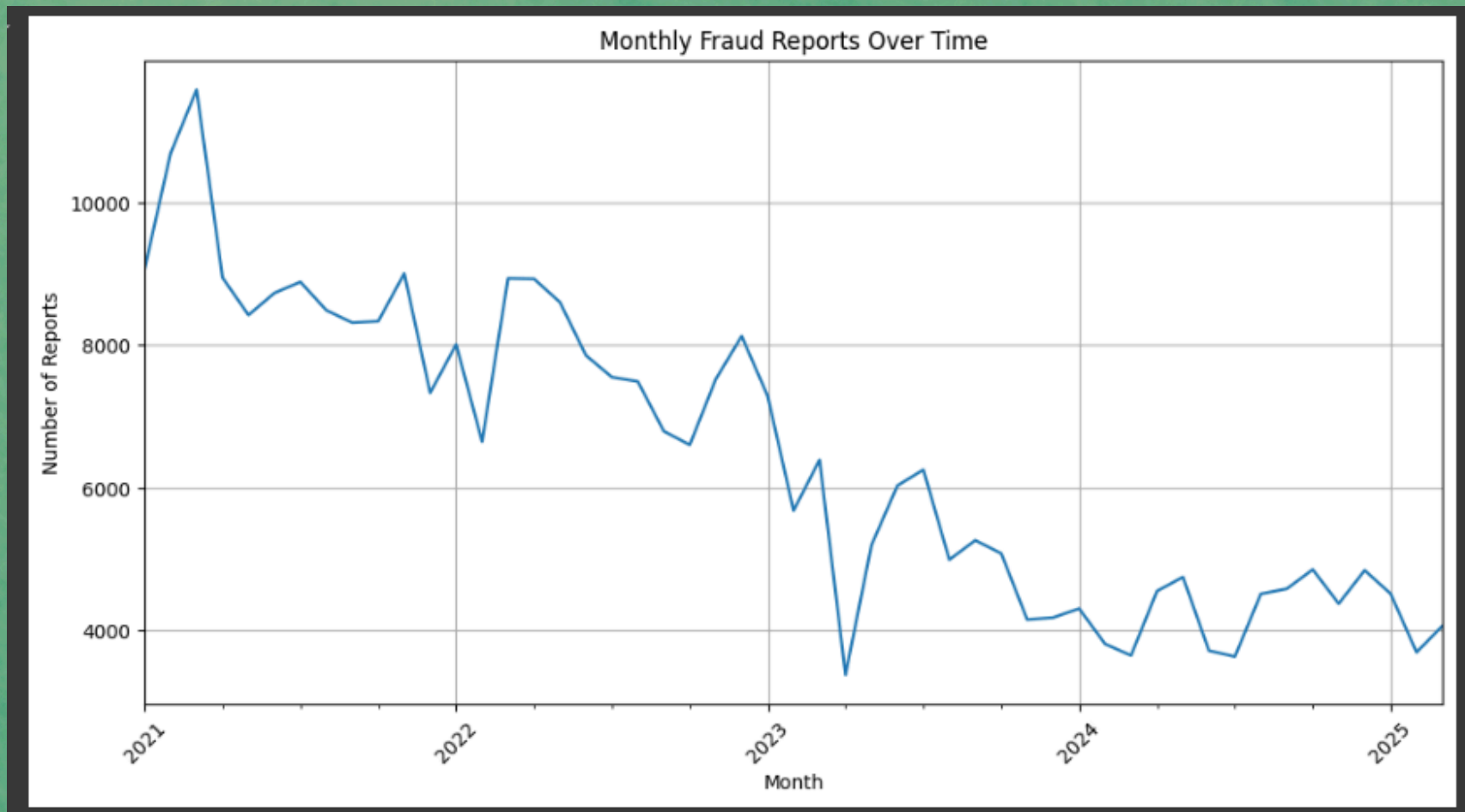
	0
ID	0
Date_Received	0
Complaint_Source	0
Country	74367
Province	77767
Fraud_Category	0
Solicitation Method	17387
Gender	84113
Language of Correspondence	84555
Victim_Age_Range	98111
Complaint Type	0
Victim_Count	0
Dollar_Loss	0

dtype: int64

```
[ ] 1 missing_summary = df.isna().sum().sort_values(ascending=False)
    2 print("\nMissing values summary:\n", missing_summary[missing_summary > 0])
```

```
Missing values summary:
Victim_Age_Range      98111
Language of Correspondence 84555
Gender                84113
Province              77767
Country               74367
Solicitation Method   17387
dtype: int64
```

Exploratory Data Analysis

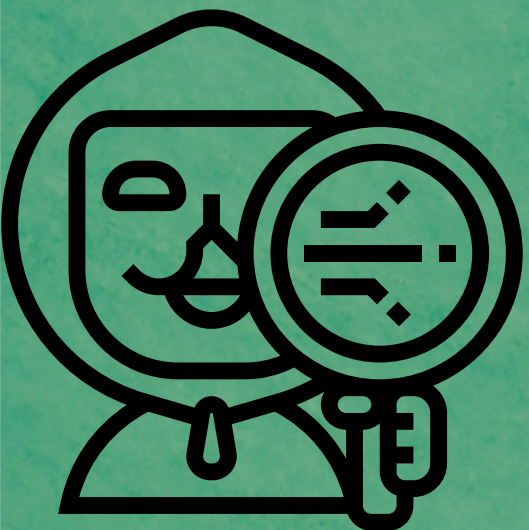


How is Financial Fraud Changing Over Time?

The line chart shows a steady decline in reported financial fraud cases from 2021 to early 2025.

- Reports peaked in early 2021 at over 11,000 per month.
- By 2024, the number of monthly reports stabilized at a much lower level, around 4,000 to 5,000 cases.

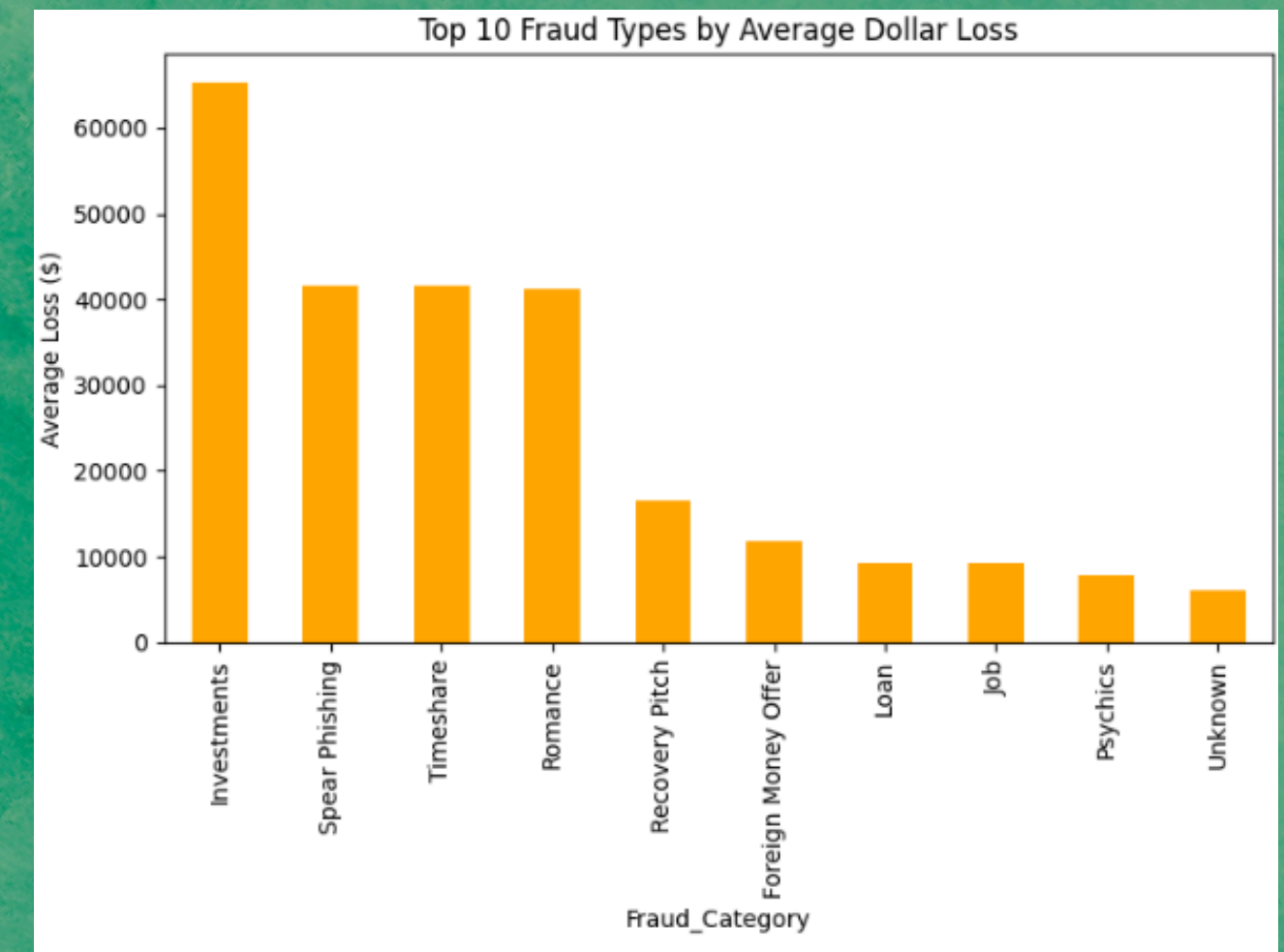
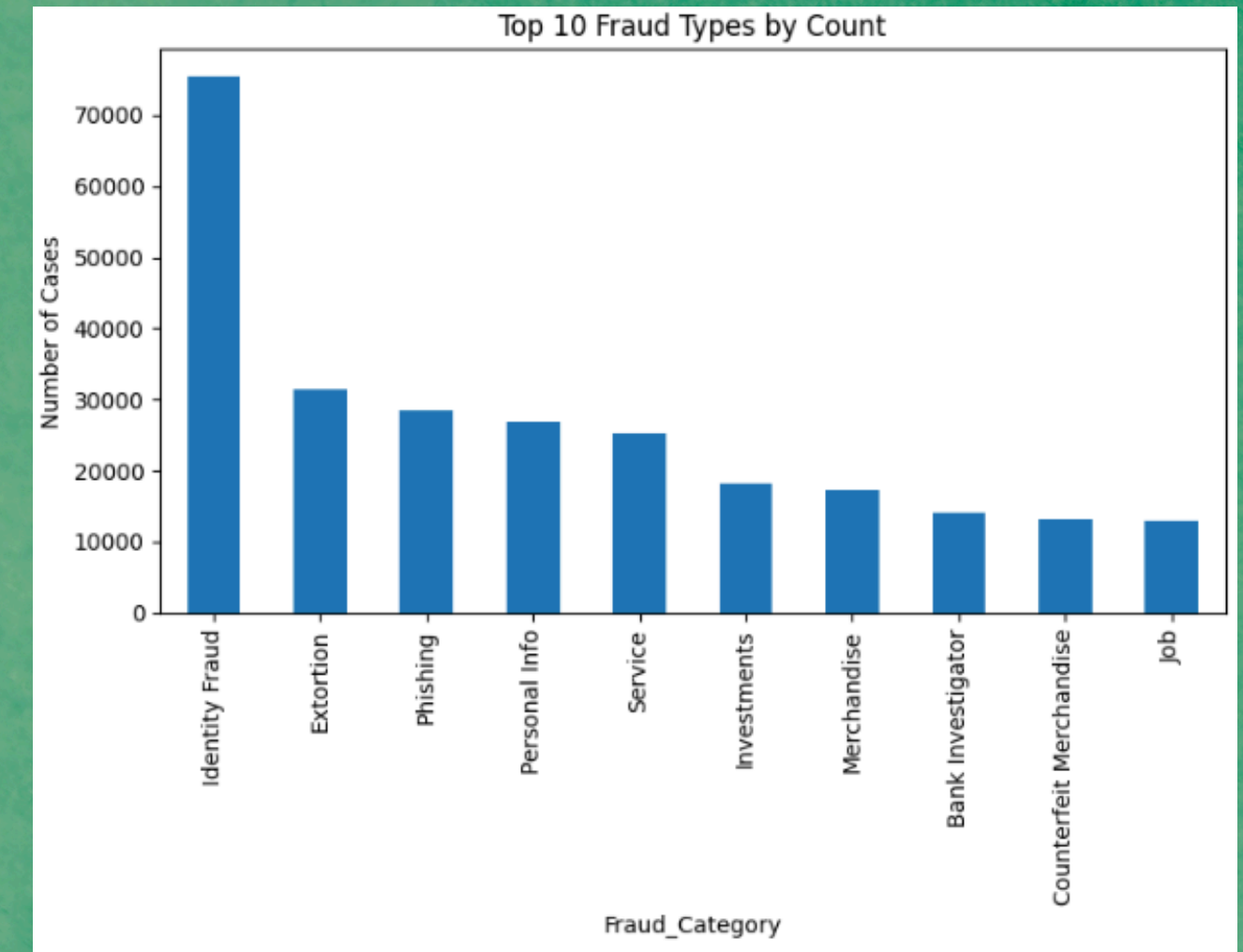
This decline may reflect increased public awareness, better fraud prevention strategies, or changes in reporting behavior.



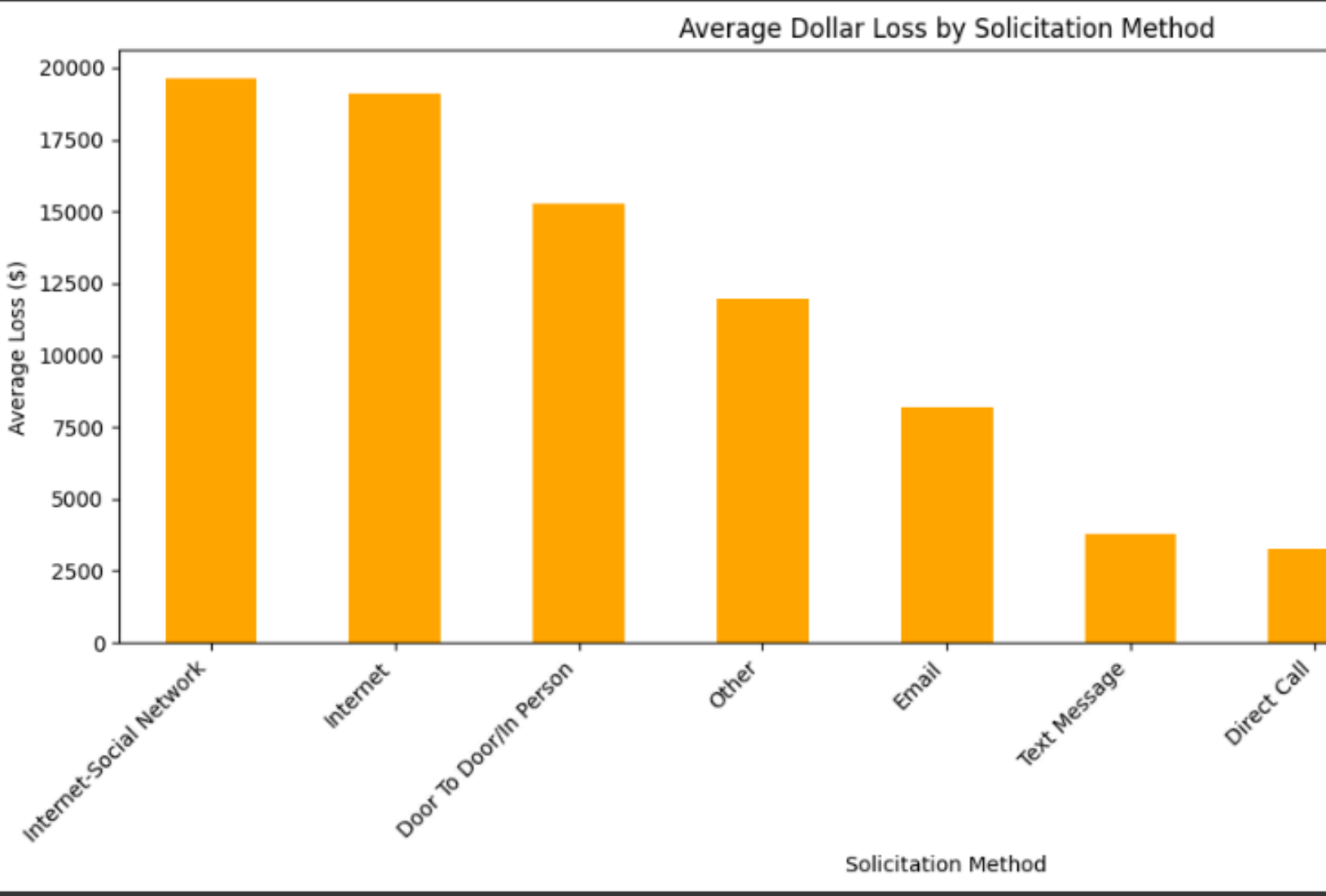
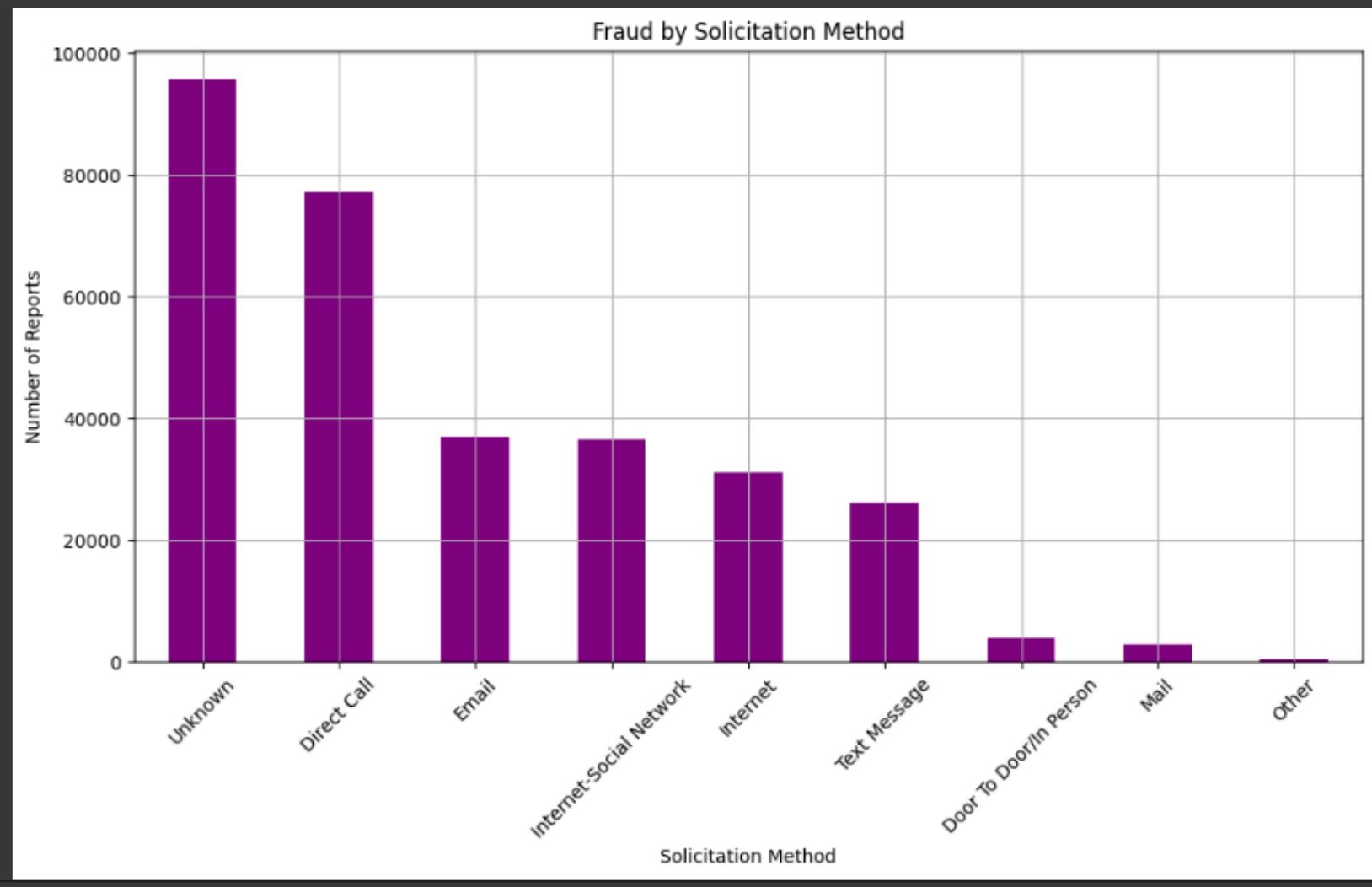
Top Fraud Categories

Frequent scams aren't always the most damaging: high-volume frauds tend to have lower average losses.

Low-frequency scams like Investment and Romance frauds, although less common, cause severe financial harm per victim. This calls for targeted awareness and prevention efforts.



Solicitation Methods by Frequency

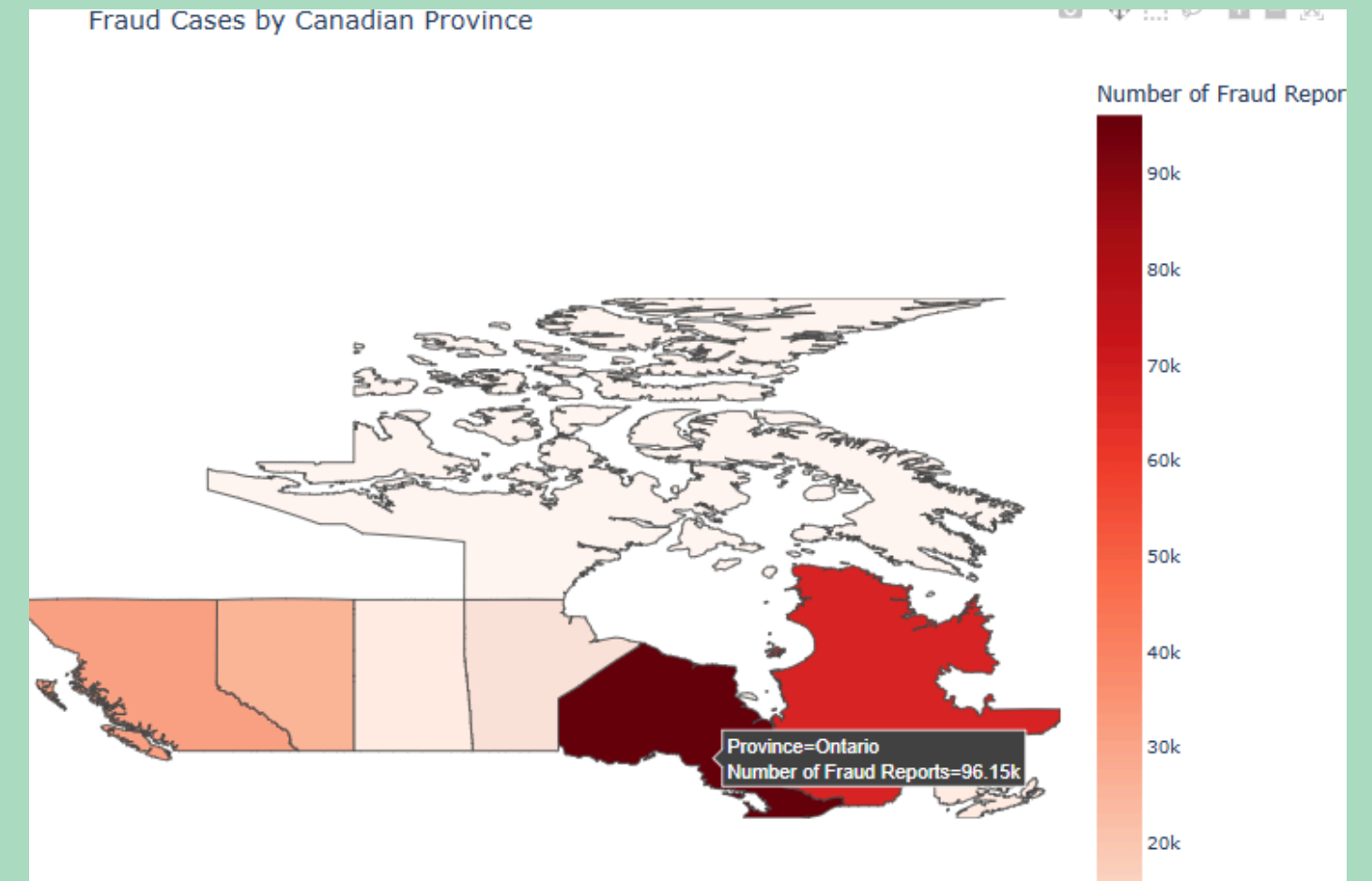
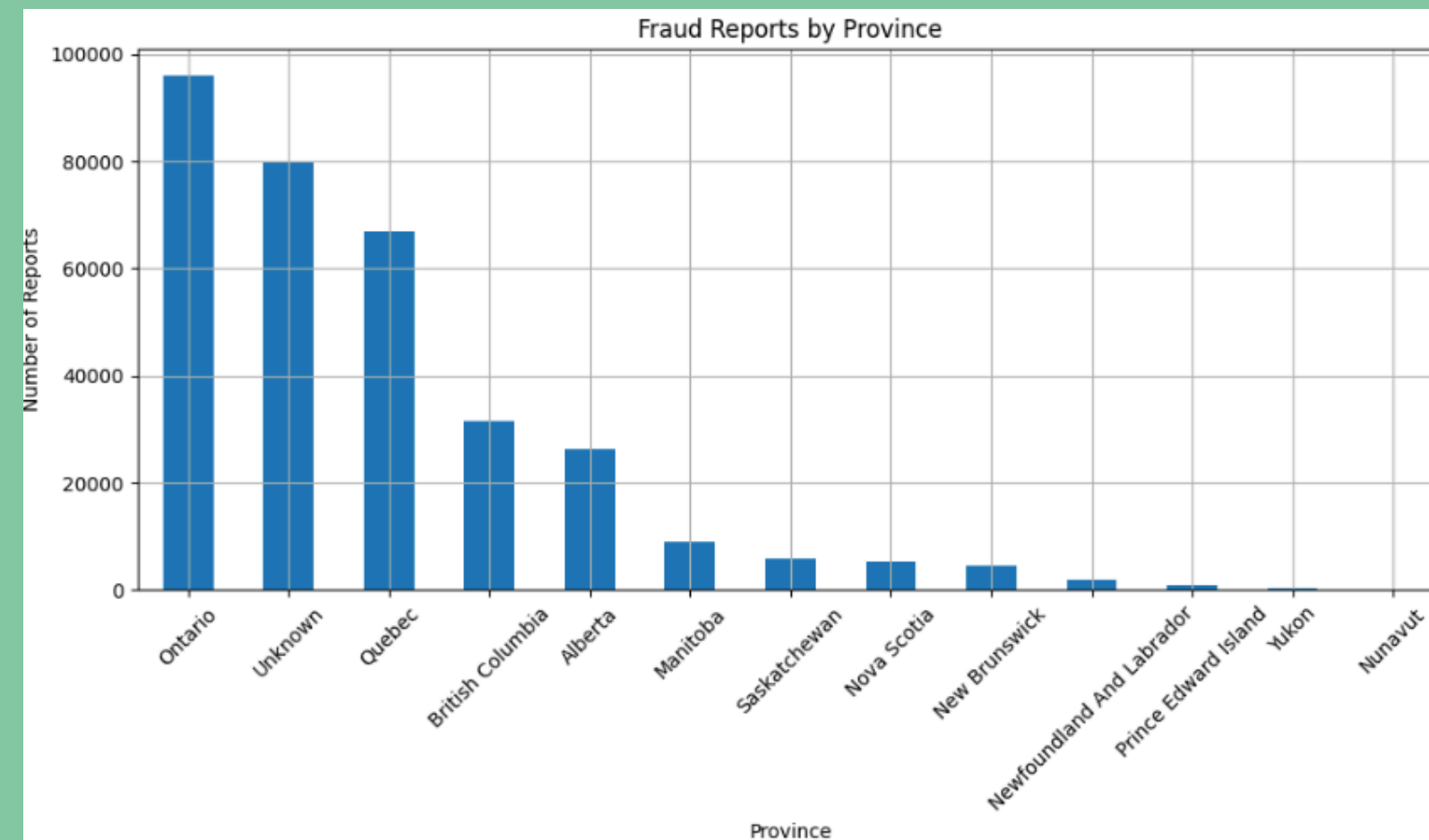


The most frequent solicitation method is Direct Call, followed by Email and Internet-based methods. However, Email and Social Media tend to result in higher average dollar losses, suggesting more successful scams.

A large portion of records is still classified as Other/unknown, indicating room for improving fraud categorization and reporting.

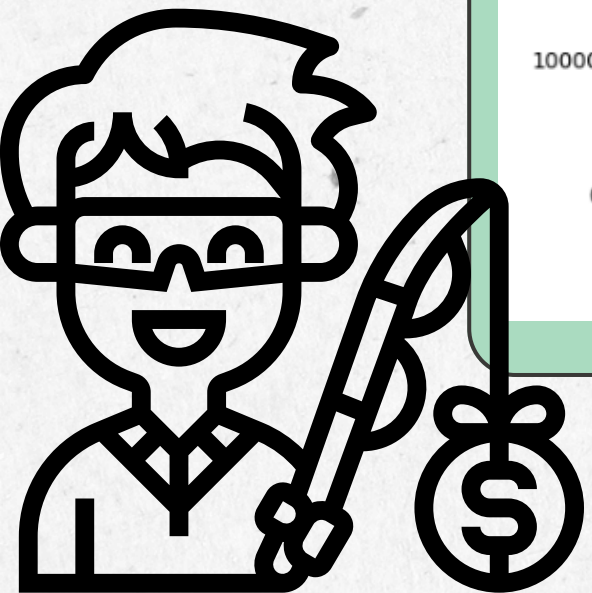
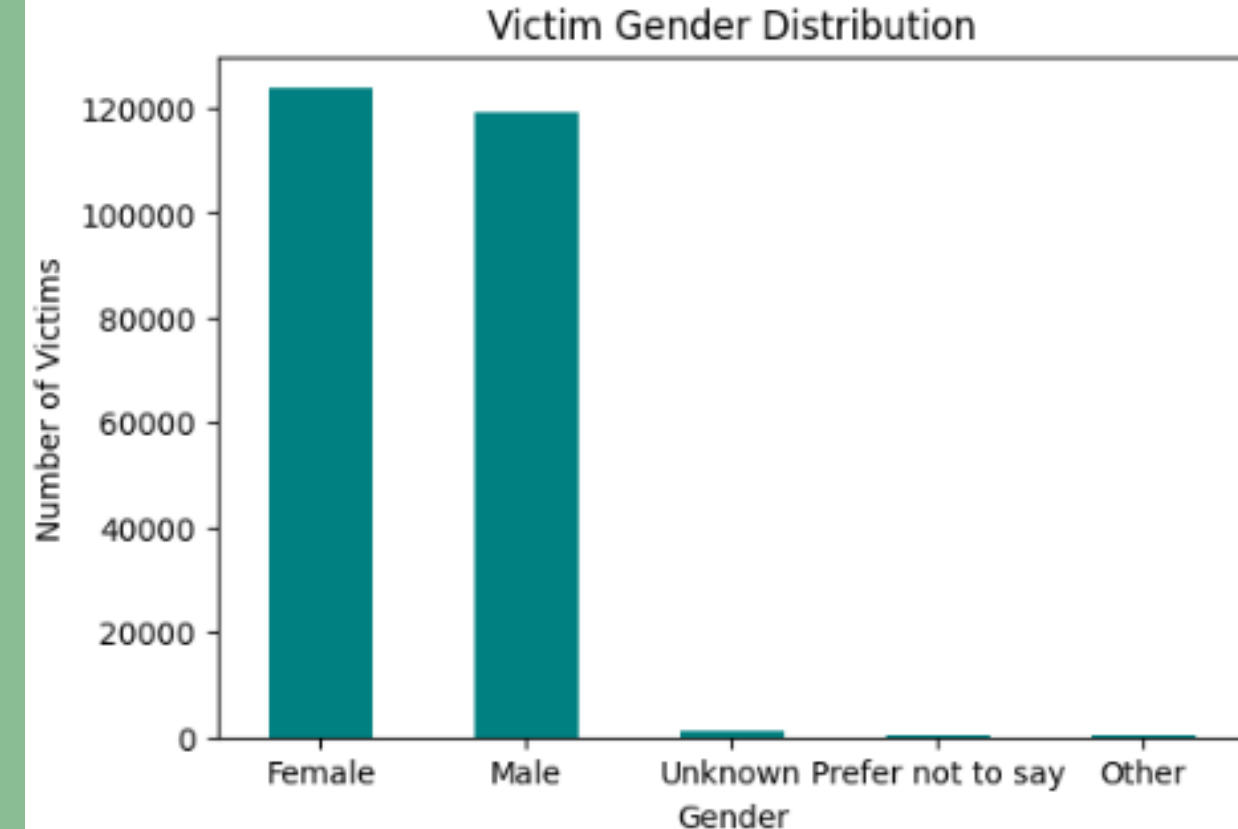
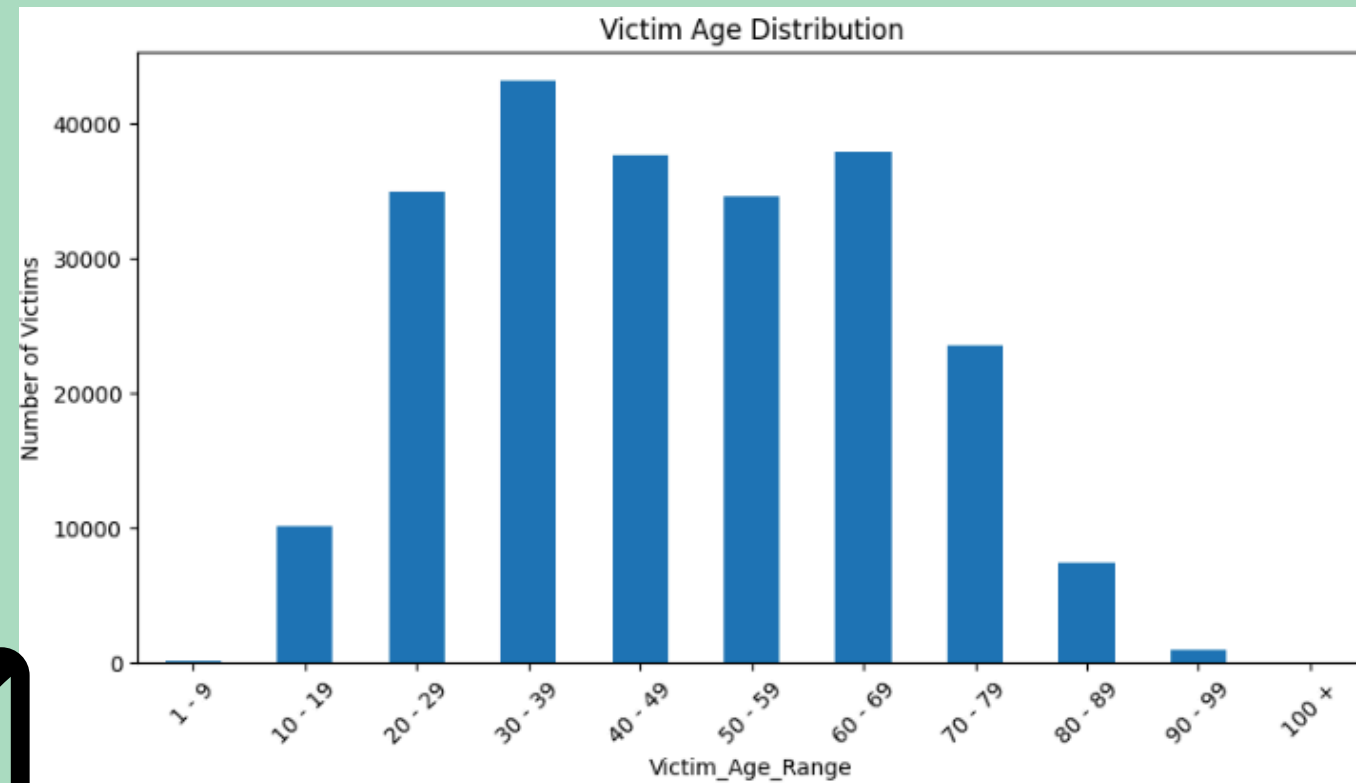
Regional Distribution of Fraud in Canada

Fraud activity appears to be more concentrated in the most populous provinces, likely due to higher population density and more digital/financial activity.



Demographics Distribution

- Adults aged 30–69 are the most frequently targeted group, likely due to higher financial activity and online presence.
- Young adults (20–29) and seniors (60–79) are also major targets.
- Very few cases in children (1–9) and centenarians (100+).
- This suggests a need for fraud prevention messaging tailored to working-age adults and seniors, with extra attention to digital literacy for youth and elderly groups.



Predictive Modeling

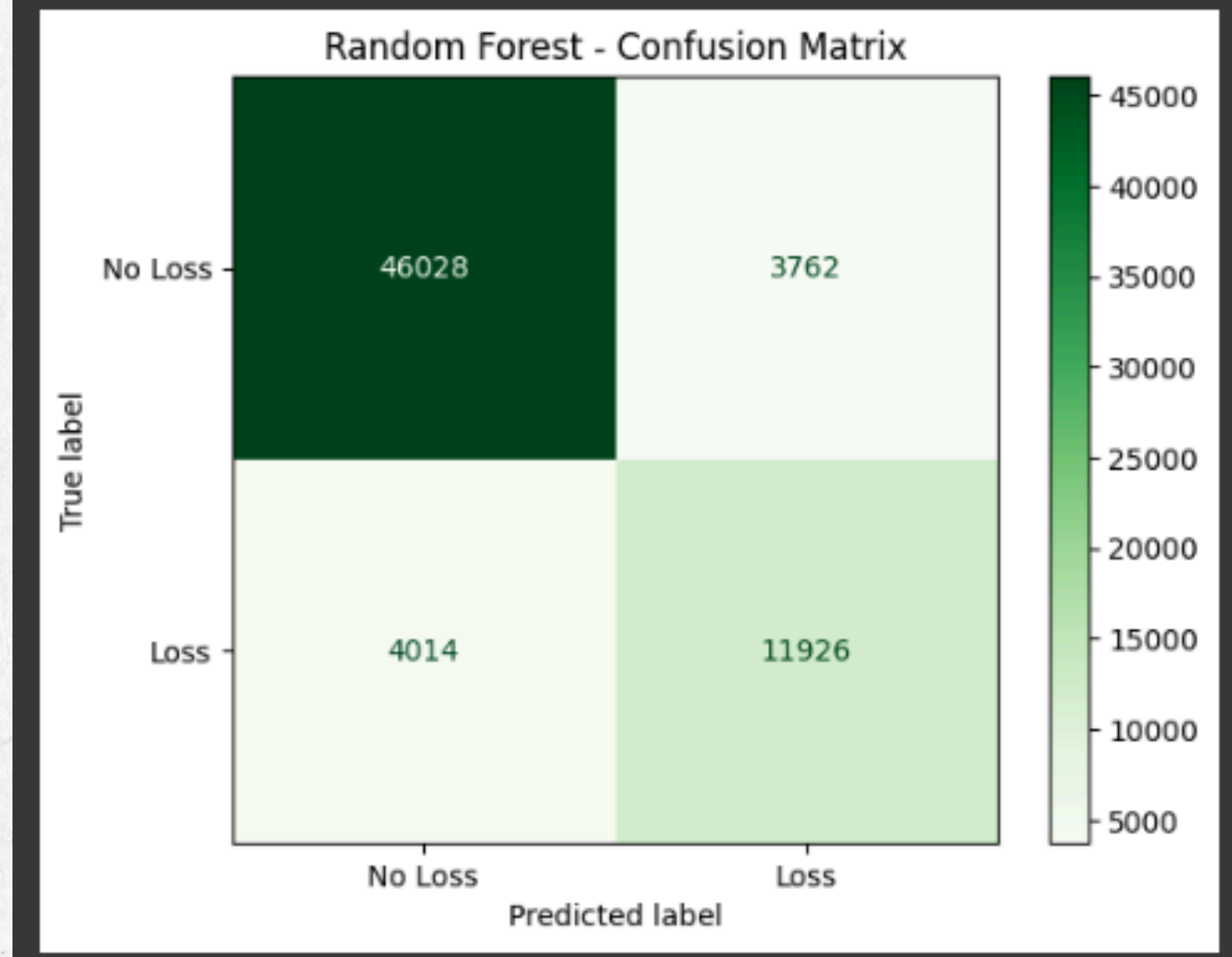
Binary classification model to predict if a fraud complaint leads to financial loss (Dollar_Loss > 0).

This model helps identify fraud cases that are more likely to result in financial damage, supporting fraud detection and prevention strategies.

Model	Accuracy	Fraud Recall	Fraud Precision	Fraud F1-Score
Random Forest	0.88	0.75	0.76	0.75
Decision Tree	0.88	0.73	0.76	0.74
Logistic Reg.	0.88	0.72	0.76	0.74

Random Forest Classification Report:

	precision	recall	f1-score	support
0	0.92	0.92	0.92	49790
1	0.76	0.75	0.75	15940
accuracy			0.88	65730
macro avg	0.84	0.84	0.84	65730
weighted avg	0.88	0.88	0.88	65730





Thank you
very much!